

STAT 341: Tutorial Test 2 - F19

First name

Last name

Signature

ID Number

Email

Time allowed: 50 minutes

Special instructions

- All numeric answers should be written either as a number to three decimal places or as a fraction reduced to its lowest terms.
- If working is not specifically requested, a correct answer will receive full marks. However, an incorrect answer could still receive credit if working is provided. An incorrect answer with no working will receive zero marks.

1. [1 mark] Provide one reason as to why probabilistic sampling is superior to non-probabilistic sampling.

Probabilistic sampling allows us to make probability statements about the sample estimate of the attribute, $a(S)$. //

2. [2 marks] Consider taking a simple random sample of size n with replacement from a population of N units. Derive the joint probability of inclusion.

Suppose $u \neq v$. $P(u \in S, v \in S)$

ASIDE:

$$P(u \in S) = 1 - \left(\frac{N-1}{N}\right)^n$$

if sampled with replacement.

$$= 1 - P(u \notin S \cup v \notin S)$$

$$= 1 - \{P(u \notin S) + P(v \notin S) - P(u \notin S, v \notin S)\}$$

$$= 1 - \left\{ \left(\frac{N-1}{N}\right)^n + \left(\frac{N-1}{N}\right)^n - \left(\frac{N-2}{N}\right)^n \right\}$$

$$\text{If } u=v, \text{ then } P(u \in S, u \in S) = P(u \in S) = 1 - \left(\frac{N-1}{N}\right)^n //$$

3. [3 marks] Consider the population attribute $\alpha(P)$. Based on a random sample S , the population attribute is estimated by the sample attribute $\alpha(S) = \tilde{\alpha}(P)$. Show that

$$MSE(\alpha(S)) = Var(\alpha(S)) + [Bias(\alpha(S))]^2$$

$$Var(\alpha(S)) = E[\alpha(S)^2] - E[\alpha(S)]^2$$

$$Bias(\alpha(S))^2 = (E[\alpha(S)] - \alpha(P))^2$$

$$= E[\alpha(S)]^2 - 2\alpha(P)E[\alpha(S)] + \alpha(P)^2$$

$$\Rightarrow Var(\alpha(S)) + Bias(\alpha(S))^2 = E[\alpha(S)^2] - 2\alpha(P)E[\alpha(S)] + \alpha(P)^2$$

$$= E[\alpha(S)^2 - 2\alpha(P)\alpha(S) + \alpha(P)^2]$$

$$= E[(\alpha(S) - \alpha(P))^2]$$

$$= MSE(\alpha(S)) \quad \square //$$

4. (a) [3 marks] Show that the HT estimator $\sum_{u \in S} y_u / \pi_u$ is unbiased for the population total. Clearly define any notation used.

$$\text{Let } I(x) = \begin{cases} 1 & \text{if unit } x \text{ is in } S \\ 0 & \text{else} \end{cases}$$

Then we have $\sum_{u \in S} y_u / \pi_u = \sum_{u \in P} I(u) y_u / \pi_u$, $P = \text{population}$

Now we can take expectation.

$$\begin{aligned} E\left[\sum_{u \in S} y_u / \pi_u\right] &= \sum_{u \in P} E[I(u)] y_u / \pi_u \\ &= \sum_{u \in P} \pi_u \cdot y_u / \pi_u \quad \text{since } E[I(u)] = \pi_u \\ &= \sum_{u \in P} y_u \\ &= \text{population total} \quad \square \end{aligned}$$

//

- (b) [2 marks] Based on the sample S , write down an estimate of the variance of the Horvitz-Thompson estimator $a_{HT}(S)$,

$$\text{Var}(a_{HT}(S)) = \sum_{u \in P} \sum_{v \in P} (\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}.$$

$$\text{Let } Z_{uv} = (\pi_{uv} - \pi_u \pi_v) \frac{y_u}{\pi_u} \frac{y_v}{\pi_v}.$$

$$\Rightarrow \text{Var}(a_{HT}(S)) = \sum_{u, v \in P} Z_{uv}, \text{ which is a total quantity.}$$

Hence, we can HT-estimate it too.

$$\hat{\text{Var}}_{HT}(a_{HT}(S)) = \sum_{u, v \in S} Z_{uv} / \pi_{uv} \quad //$$

5. A sample ($n = 6$) was randomly selected from a population ($N = 124$). The inclusion probabilities and the observed y values from the sample are

y_i	-2	-1	0	1	2	3
π_i	4/50	4/50	4/50	4/50	1/73	1

- (a) [2 marks] Give the HT estimate for the proportion of units less than or equal to -2.

$$\hat{P}_{HT}(Y \leq -2) = \frac{1}{N} \sum_{i=1}^6 I(y_i \leq -2) / \pi_i = \frac{1}{124} \cdot 1 / (4/50) = 0.1008 \quad (\approx 0.101) //$$

- (b) [2 marks] Give the HT estimate for the population average.

$$\begin{aligned} \bar{y}_{HT} &= \frac{1}{124} \sum_{i=1}^6 y_i / \pi_i \\ &= \frac{1}{124} \left(-2 / (4/50) + -1 / (4/50) + 0 + 1 / (4/50) + 2 / (1/73) + 3 / 1 \right) \\ &= 1 // \end{aligned}$$

- (c) [2 marks] Give the HT estimate of the population median.

$$\begin{aligned} \hat{P}_{HT}(Y \leq 1) &= \frac{1}{124} \left(1 / (4/50) + 1 / (4/50) + 1 / (4/50) + 1 / (4/50) \right) \\ &= \frac{50}{124} < 0.5 \end{aligned}$$

$$\hat{P}_{HT}(Y \leq 2) = \frac{1}{124} (50 + 73) = \frac{123}{124} > 0.5$$

$$\text{Since } \text{median} = \inf \{ y \in P : P(Y \leq y) \geq 0.5 \},$$

$$\hat{\text{median}}_{HT} = 2 //$$

6. (a) [4 marks] Describe in point form a test of significance using a discrepancy measure $D(P_1, P_2)$

1. Generate a hypothesis: P_1 & P_2 are drawn from the same population.
2. Construct a discrepancy measure $D(P_1, P_2)$, where large values indicate evidence against the hypothesis.
3. Compute d , the observed discrepancy measure.
4. Compute the probability of $D \geq d$ when the hypothesis is true. //

(b) [1 mark] What is the observed significance level?

The probability that two sets of sample have a larger discrepancy value than the observed discrepancy, d , given that the hypothesis is true. //

ie, $P(D \geq d \mid \text{hypothesis is true})$

7. [6 marks (1 mark each)] Determine whether the following statements are true or false.

(a) When sampling with replacement the sample average is a Horvitz-Thompson estimate.

i. True

ii. False

(b) The Horvitz-Thompson estimator of the population mean is a scale equivariant estimator.

i. True

ii. False

(c) The difference between the attribute of the study population and that of the sample is called the study error.

i. True

ii. False

(d) The observed significance level measures evidence in favour of the hypothesis.

i. True

ii. False

(e) The relative error of an estimator is a number between 0 and 1.

i. True

ii. False

(f) We can quantify the sampling error by calculating all possible samples.

i. True

ii. False